15A, 600V - 800V Low V_F Standard Bridge Rectifier

FEATURES

- AEC-Q101 qualified available
- Low forward drop enhance the efficiency
- Oxide planar chip junction
- High surge current capability
- UL Recognized File # E-326243
- RoHS Compliant
- Halogen-free according to IEC 61249-2-21

APPLICATIONS

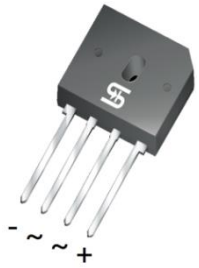
- Switching mode power supply (SMPS)
- Adapters
- Lighting application

MECHANICAL DATA

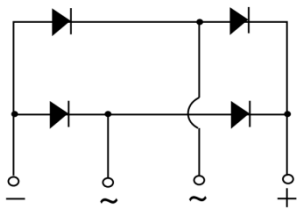
- Case: GBU
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Polarity: As marked
- Weight: 4.00g (approximately)

KEY PARAMETERS		
PARAMETER	**VALUE**	**UNIT**
I_F	15	A
V_{RRM}	600 - 800	V
I_{FSM}	200	A
$T_{J\ MAX}$	150	°C
Package	GBU	
Configuration	Quad	

Pb-Free RoHS COMPLIANT HALOGEN FREE



GBU



ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)					
PARAMETER		**SYMBOL**	**GBU15L05**	**GBU15L06**	**UNIT**
Marking code on the device			GBU15L05	GBU15L06	
Repetitive peak reverse voltage		V_{RRM}	600	800	V
Reverse voltage, total rms value		$V_{R(RMS)}$	420	560	V
Forward current		I_F	15		A
Surge peak forward current single half sine-wave superimposed on rated load	t = 8.3ms	I_{FSM}	200		A
	t = 1.0ms		630		A
Rating for fusing (t<8.3ms)		I^2t	166		A^2s
Junction temperature		T_J	- 55 to +150		°C
Storage temperature		T_{STG}	- 55 to +150		°C

THERMAL PERFORMANCE

PARAMETER	SYMBOL	TYP	UNIT
Junction-to-ambient thermal resistance	$R_{\Theta JA}$	15	°C/W
Junction-to-case thermal resistance	$R_{\Theta JC}$	3	°C/W

ELECTRICAL SPECIFICATIONS (T_A = 25°C unless otherwise noted)

PARAMETER		CONDITIONS	SYMBOL	TYP	MAX	UNIT
Forward voltage per diode(1)	GBU15L05	I_F = 7.5A, T_J = 25°C	V_F	0.87	0.90	V
		I_F = 7.5A, T_J = 125°C		0.75	-	V
	GBU15L06	I_F = 7.5A, T_J = 25°C		0.93	0.96	V
		I_F = 7.5A, T_J = 125°C		-	-	V
Reverse current @ rated V_R per diode(2)		T_J = 25°C	I_R	-	5	μA
		T_J = 125°C		-	150	μA

Notes:

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

ORDERING INFORMATION

ORDERING CODE(1)(2)	PACKAGE	PACKING
GBU15L0x	GBU	20 / Tube
GBU15L0xH	GBU	20 / Tube

Notes:

1. “x” defines voltage from 600V(GBU15L05) to 800V(GBU15L06)
2. “H” means AEC-Q101 qualified

CHARACTERISTICS CURVES

(T_A = 25°C unless otherwise noted)

Fig.1 Forward Current Derating Curve

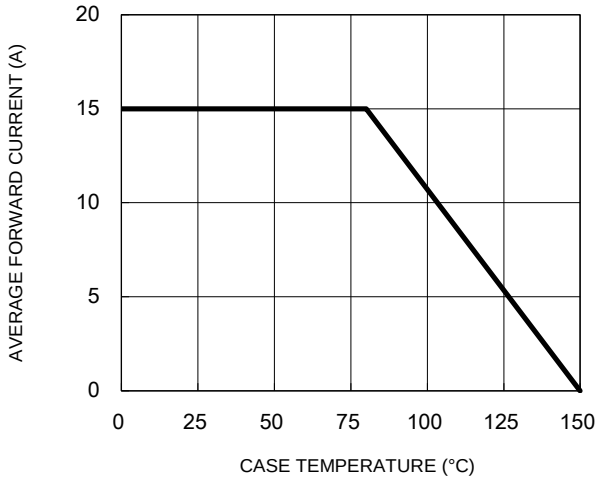


Fig.2 Typical Junction Capacitance

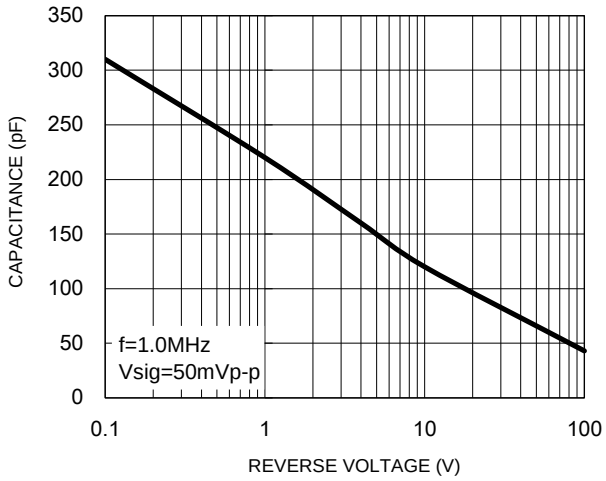


Fig.3 Typical Reverse Characteristics

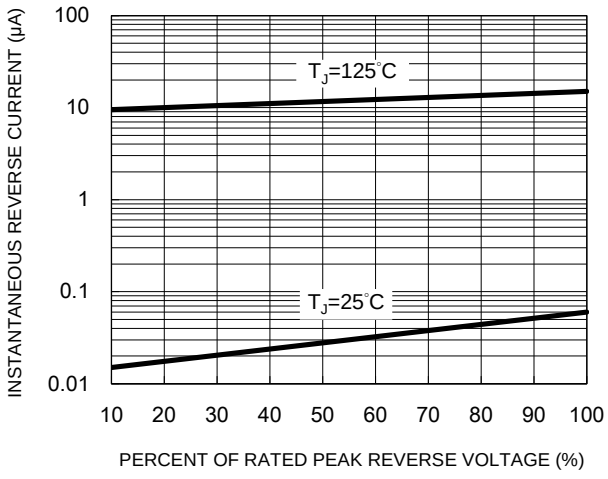


Fig.4 Typical Forward Characteristics

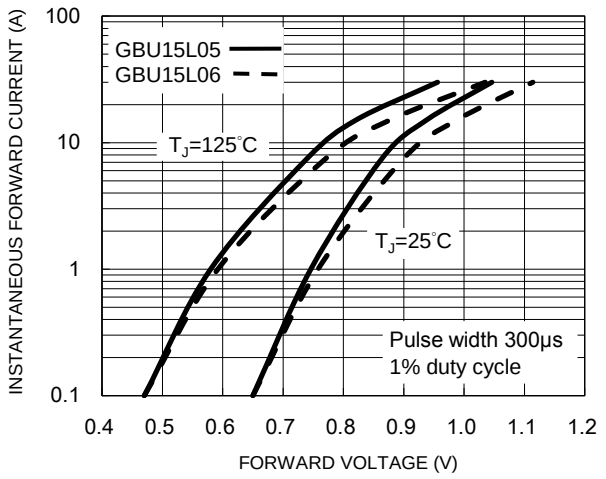
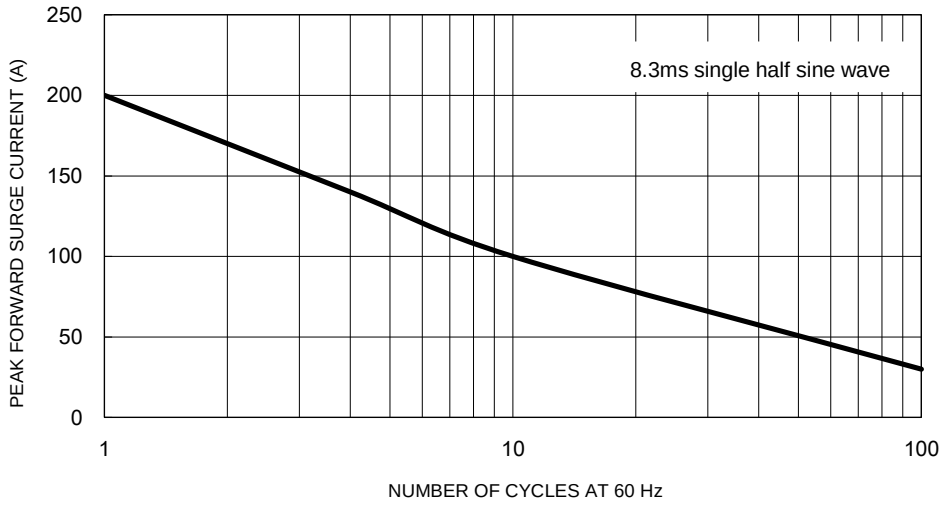
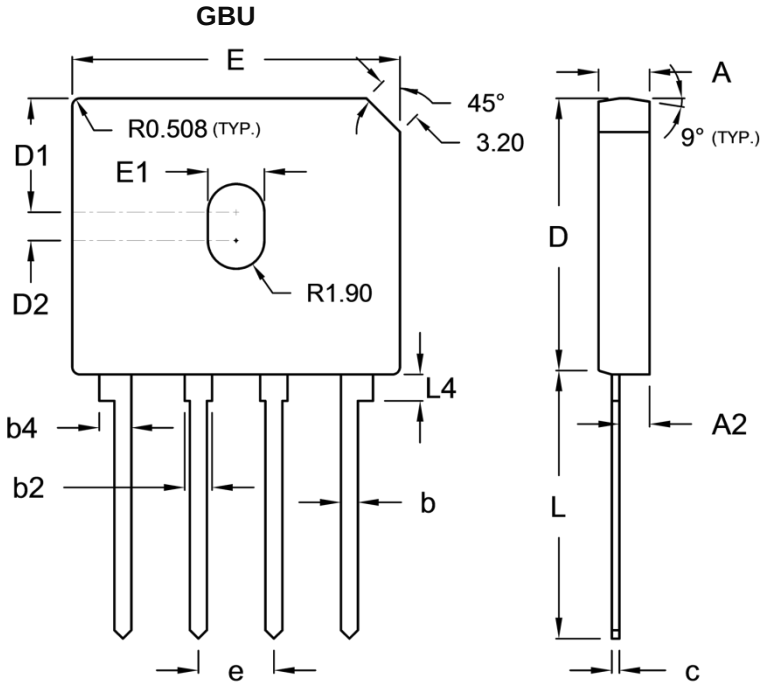


Fig.5 Maximum Non-Repetitive Forward Surge Current



PACKAGE OUTLINE DIMENSIONS



DIM.	Unit (mm)		Unit (inch)	
	Min.	Max.	Min.	Max.
A	3.30	3.56	0.130	0.140
A2	1.90	2.16	0.075	0.085
b	1.02	1.27	0.040	0.050
b2	1.65	2.03	0.065	0.080
b4	2.16	2.54	0.085	0.100
c	0.46	0.56	0.018	0.022
D	18.30	18.80	0.720	0.740
D1	7.40	7.90	0.291	0.311
D2	1.65	2.16	0.065	0.085
E	21.80	22.30	0.858	0.878
E1	3.50	4.10	0.138	0.161
e	4.83	5.33	0.190	0.210
L	17.50	18.00	0.689	0.709
L4	1.52	2.03	0.060	0.080

MARKING DIAGRAM

P/N = Marking Code
G = Green Compound
YWW = Date Code
F = Factory Code

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